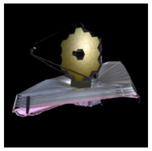

1883 - A Modern Theory for Strong Line Metallicity Indicators in the Era of JWST

Cycle: 1, Proposal Category: AR

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OBSERVATIONS

ABSTRACT

The evolution of galaxy metallicities over cosmic time encodes the key processes encompassed in the physics of baryon cycling. NIRSpec on JWST will transform our understanding of galaxy metallicities at z>2 via the routine detection of strong (rest frame optical) nebular emission lines. And yet, there is substantial evidence that our current strong line metallicity indicators that have been calibrated for physical conditions at z~0 do not apply at high redshift. We propose to develop the most advanced theoretical strong line metallicity calibrations to date in an effort to aid the interpretation of the vast suite of near infrared emission lines that will be detected by JWST. Our models will derive the physical conditions of high redshift galaxies from modern cosmological simulations, and compute, for the first time ever, their expected emission line signal from a diverse range of sources. Our simulations include emission from HII regions on a particle by particle basis, AGN, and diffuse ionized gas, and will investigate the impact of non Solar abundance patterns, rotating stars, and binary stars on nebular emission lines. These proposed models will significantly advance theoretical strong line calibrations from the single zone photoionization models that are the current status quo, and bring metallicity indicators into

the modern era.

OBSERVING DESCRIPTION

N/A